

Bernoulli Equation Homework

Student Version

Fluid Dynamics

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Use SI units unless otherwise stated. Take

$$g = 9.81 \text{ m/s}^2, \quad \rho_{\text{water}} = 1000 \text{ kg/m}^3.$$

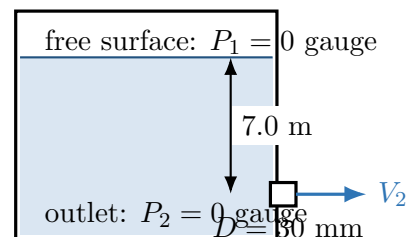
Neglect friction losses unless the problem states otherwise. Use gauge pressure consistently.

Homework 1: Water Discharging from an Open Tank

Problem

A large open tank contains water. A small circular outlet with diameter $D = 30 \text{ mm}$ is located 7.0 m below the free surface. Neglect losses and determine:

1. the ideal exit velocity;
2. the volume flow rate through the outlet;
3. the flow rate in liters per second.

Real setup**Engineering diagram**

Open tank, water, elevation difference $z_1 - z_2 = 7.0 \text{ m}$, outlet diameter $D = 30 \text{ mm}$.

Hint

Choose point 1 at the free surface and point 2 at the outlet. For a large tank, $V_1 \approx 0$, and both points are exposed to atmospheric pressure. Use

$$V_2 = \sqrt{2g(z_1 - z_2)}, \quad Q = A_2 V_2, \quad A_2 = \frac{\pi D^2}{4}.$$

Solution

Apply Bernoulli between the free surface and the outlet:

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2.$$

Since $P_1 = P_2$ and $V_1 \approx 0$,

$$V_2 = \sqrt{2g(z_1 - z_2)} = \sqrt{2(9.81)(7.0)} = 11.72 \text{ m/s}.$$

The outlet area is

$$A_2 = \frac{\pi(0.030)^2}{4} = 7.069 \times 10^{-4} \text{ m}^2.$$

Therefore,

$$Q = A_2 V_2 = (7.069 \times 10^{-4})(11.72) = 8.28 \times 10^{-3} \text{ m}^3/\text{s}.$$

Using $1 \text{ m}^3 = 1000 \text{ L}$,

$$Q = 8.28 \text{ L/s}.$$

$$V_2 = 11.72 \text{ m/s}, \quad Q = 8.28 \times 10^{-3} \text{ m}^3/\text{s} = 8.28 \text{ L/s}$$

Homework 2: Horizontal Pipe Reducer

Problem

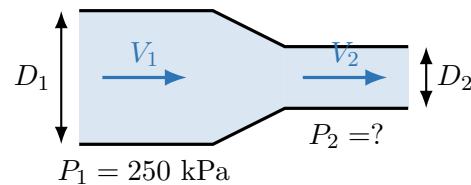
Water flows steadily through a horizontal reducer. The volume flow rate is $Q = 0.018 \text{ m}^3/\text{s}$. At section 1, the diameter is $D_1 = 0.12 \text{ m}$ and the gauge pressure is $P_1 = 250 \text{ kPa}$. At section 2, the diameter is $D_2 = 0.060 \text{ m}$. Determine:

1. the mean velocities V_1 and V_2 ;
2. the gauge pressure P_2 .

Real setup



Engineering diagram



Horizontal water flow: $Q = 0.018 \text{ m}^3/\text{s}$, $D_1 = 0.12 \text{ m}$, $D_2 = 0.060 \text{ m}$,
 $P_1 = 250 \text{ kPa gauge}$.

Hint

Use

$$V_1 = \frac{Q}{A_1}, \quad V_2 = \frac{Q}{A_2}, \quad A = \frac{\pi D^2}{4}.$$

Because the pipe is horizontal, $z_1 = z_2$. Then apply

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2.$$

Solution

The cross-sectional areas are

$$A_1 = \frac{\pi(0.12)^2}{4} = 0.01131 \text{ m}^2, \quad A_2 = \frac{\pi(0.060)^2}{4} = 0.002827 \text{ m}^2.$$

Thus,

$$V_1 = \frac{0.018}{0.01131} = 1.59 \text{ m/s}, \quad V_2 = \frac{0.018}{0.002827} = 6.37 \text{ m/s}.$$

For a horizontal reducer,

$$P_2 = P_1 + \frac{1}{2}\rho (V_1^2 - V_2^2).$$

Substitution gives

$$P_2 = 250000 + \frac{1}{2}(1000) [(1.59)^2 - (6.37)^2] = 2.31 \times 10^5 \text{ Pa}.$$

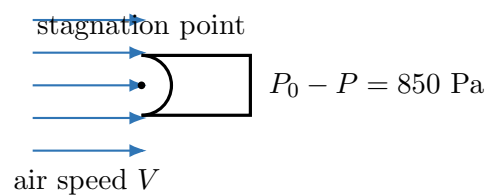
Therefore,

$$P_2 \approx 231 \text{ kPa gauge}.$$

$$V_1 = 1.59 \text{ m/s}, \quad V_2 = 6.37 \text{ m/s}, \quad P_2 = 231 \text{ kPa gauge}$$

Homework 3: Aircraft Pitot-Tube Measurement**Problem**

A Pitot tube mounted on an aircraft measures a stagnation pressure that is 850 Pa greater than the static pressure. The local air density is 1.18 kg/m^3 . Neglect elevation change and determine the airspeed.

Real setup**Engineering diagram**

Pitot measurement: $P_0 - P = 850 \text{ Pa}$, $\rho_{air} = 1.18 \text{ kg/m}^3$.

Hint

Use the stagnation-pressure relation

$$P_0 - P = \frac{1}{2} \rho V^2.$$

Solve directly for V .

Homework 4: Pressurized Oil Tank

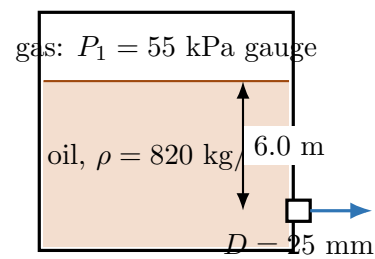
Problem

A large tank contains oil with density $\rho = 820 \text{ kg/m}^3$. The gas above the oil is maintained at a gauge pressure of 55 kPa. A 25 mm-diameter outlet is located 6.0 m below the free surface and discharges to the atmosphere. Neglect losses and determine the ideal exit velocity and volume flow rate.

Real setup



Engineering diagram



Oil tank: $\rho = 820 \text{ kg/m}^3$, $P_1 = 55 \text{ kPa gauge}$, $z_1 - z_2 = 6.0 \text{ m}$, outlet $D = 25 \text{ mm}$.

Hint

For a large tank, $V_1 \approx 0$, and the outlet pressure is zero gauge. Use

$$V_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho} + 2g(z_1 - z_2)}, \quad Q = \frac{\pi D^2}{4} V_2.$$

Homework 5: Nozzle Diameter Design

Problem

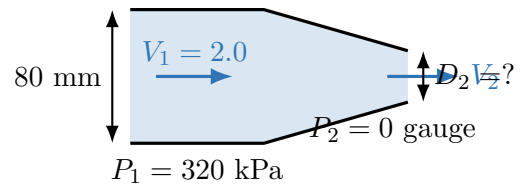
Water enters a horizontal nozzle through an 80 mm-diameter pipe at a mean velocity of 2.0 m/s and a gauge pressure of 320 kPa. The nozzle discharges to the atmosphere. Neglect losses. Determine:

1. the ideal outlet velocity;
2. the volume flow rate;
3. the required outlet diameter.

Real setup



Engineering diagram



Horizontal nozzle: $D_1 = 80$ mm, $V_1 = 2.0$ m/s, $P_1 = 320$ kPa gauge, atmospheric outlet.

Hint

First use Bernoulli to determine V_2 :

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2.$$

Then calculate $Q = A_1 V_1$, use $A_2 = Q/V_2$, and obtain

$$D_2 = \sqrt{\frac{4A_2}{\pi}}.$$

End of Bernoulli Equation Homework